Worksheet 8-3 A Definite Integrals in Applied Context #1
AP Calculus Name_____

1.) **Set up an expression or equation for each of the following questions.**
At time $t = 0$ hours, there are 213 cars in the parking lot of a large apartment complex. For $0 \leq t \leq 7$ hours, the rate that cars are entering the parking lot can be modeled by the function $E(t)$, measured in cars per hour. During this same time period, cars leave the parking lot at the rate $L(t)$, measured in cars per hour.

a) How many cars leave the parking lot during the first 4 hours?

b) How many cars enter the parking lot from time $t = 1$ to $t = 6$ hours?

c) How many cars are in the parking lot at time $t = 5$?

d) Write an equation involving one or more integrals for $A(t)$, the number of cars in the parking lot at any time t, for $0 \leq t \leq 7$.

2.) A large beef farm breeds cows and raises them until they are about seven months old, when they are sold. At time $t = 0$ months, the beef farm has 1407 cows. For $0 \leq t \leq 6$ months, the birthrate of cows on the farm can be modeled by the function

$$B(t) = 235 - 211\cos\left(\frac{\pi t}{7}\right) \qquad \text{measured in cows per month.}$$

During the same time period, cows are being sold at the rate

$$S'(t) = \frac{8013}{5t^2 - 18t + 40} \qquad \text{measured in cows per month.}$$

a) How many cows are on the farm at time $t = 4$ months?

b) Write an expression involving one or more integrals that gives the number of cows on the farm at any time t, where $0 \leq t \leq 6$ months.

3.) A factory produces brake pads for automobiles. In the factory, the finished brake pads are sent down a long conveyer belt where they are dropped into a large bin and taken for inspection. Brake pads are being placed on the conveyer belt at the rate modeled by

$$F'(t) = -0.017t^4 + 0.76t^3 - 10.85t^2 + 46.92t + 67.1 \quad \text{for } 0 \le t \le 20 \text{ minutes}$$

where $F'(t)$ is measured in brake pads per minute. During this same time interval, the brake pads are falling off the conveyer belt into the large bin at the constant rate of 60 brake pads per minute. There are no brake pads when the conveyer belts first turn on ($t = 0$ minutes).

a) After 5 minutes, how many total brake pads have been added onto the conveyer belt?

b) How many brake pads fall into the large bin from $t = 3$ to $t = 12$ minutes?

c) At time $t = 20$ minutes, how many brake pads are on the conveyer belt?

d) Write an equation for the amount of brake pads, $A(t)$, on the conveyer belt at time t minutes.

e) If there are more than 350 brake pads on the conveyer at any given time, the factory is penalized for not following standard procedures. Write an integral involving one or more integrals that, if solved, would find the times r, $0 \le r \le 20$ minutes, when the conveyer belt is over the limit of 350 brake pads.

4.) For $0 \le x \le 10, g'(x) = 1.3 \ln\left(\frac{x^2 + 3x + 7}{x + 1}\right)$. It is known that $g(3) = -12$.

a) Find $g(8)$

b) Find $g(0)$

5.) At 7AM, there are 43 employees inside a large office building. Employees enter the building at a rate modeled by

$$g(t) = 25\left|(10 - t)\sin\left(\frac{\pi t}{5}\right)\right| \text{ for } 0 \le t \le 10$$

where $g(t)$ is measured in people per hour and t is the number of hours since 7AM. Due to a company policy, from 7AM to 9AM no employees leave the office building. Starting at 9AM, employees exit the building at a rate modeled by

$$f(t) = 97 - 81\cos\left(\frac{\pi t - 2t}{2.9}\right) \text{ for } 2 \le t \le 10$$

where $f(t)$ is measured in people per hour and t is the number of hours since 7AM.

a) How many employees entered the building from 7AM to 9AM?

b) How many employees have left the building by 3PM?

c) Write an equation for $P(t)$ where $P(t)$ gives the number of employees in the building at any time t for $2 \le t \le 10$.

6.) The number of trees that are planted in a large forest is changing at the rate

$$p'(t) = -0.2(t - 11)^2 + 131$$

In the same forest, trees are being cut down at the rate

$$c'(t) = 37\ln(0.84t + 3.8)$$

where $c'(t)$ and $p'(t)$ are both measured in trees per day and t is measured in days. There are 1200 trees in the forest at time $t = 0$ days.

a) How many trees are planted during the first 15 days?

b) How many trees are in the forest after 30 days?